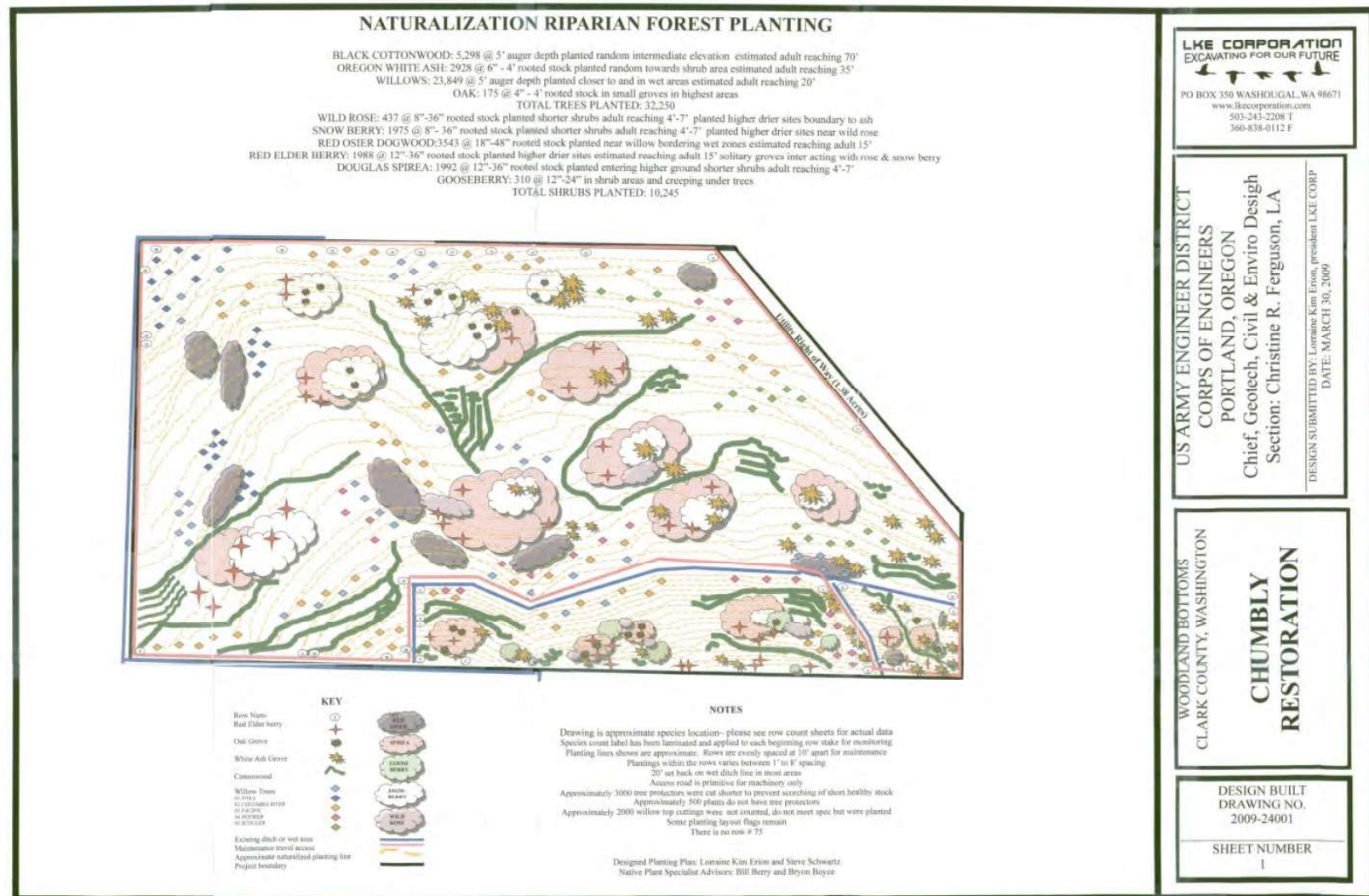


Attachment 3: Chumbley Wildlife Mitigation Site As-Built Planting Map - Original Planting 2009



 : 75 red osier

 : elderberry

 : 400 snowberries
(2/9/18)

 : 400 spirea

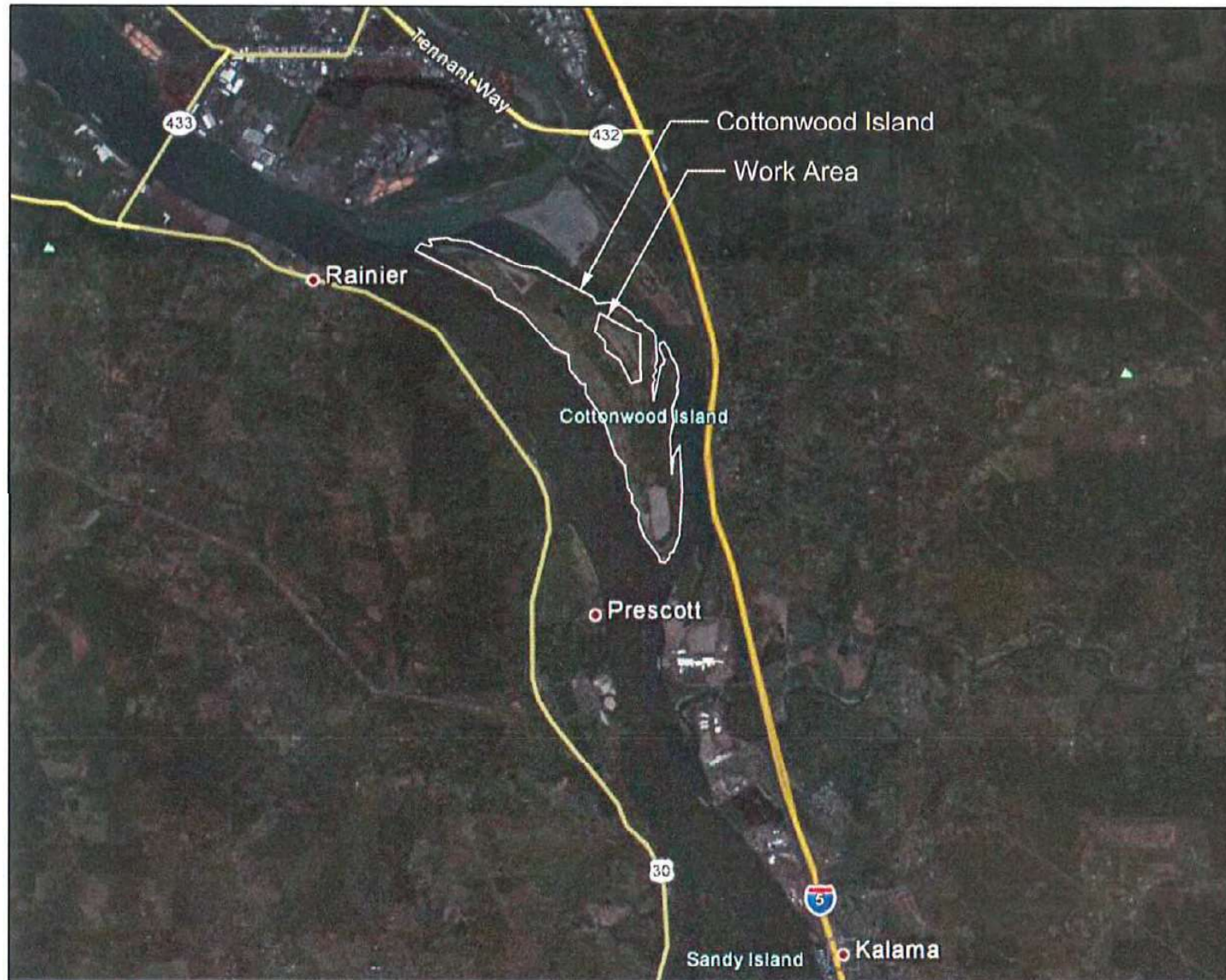
 : willow

Chumbley Wildlife Mitigation Area

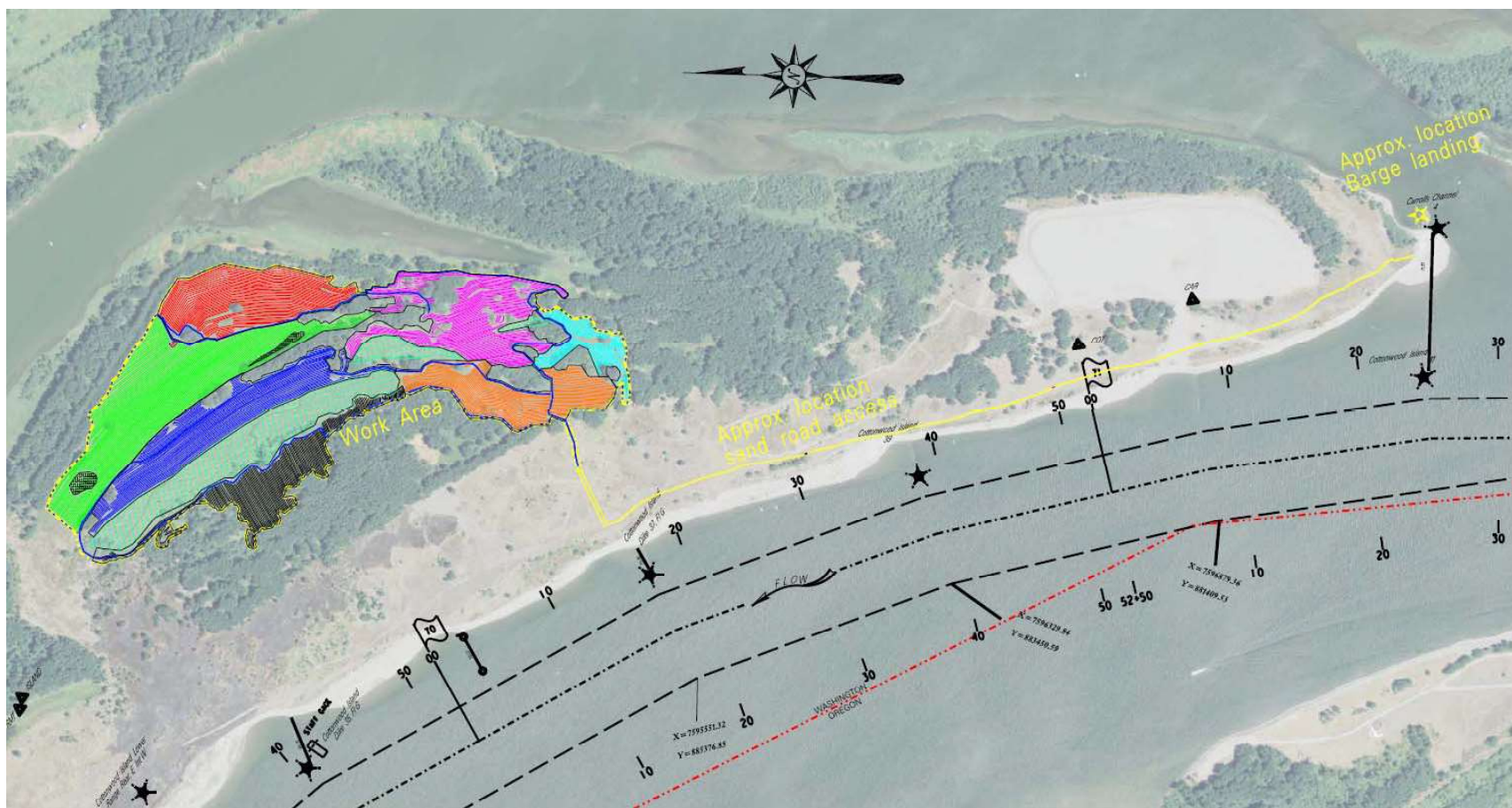
Progress Map for 02 / 12



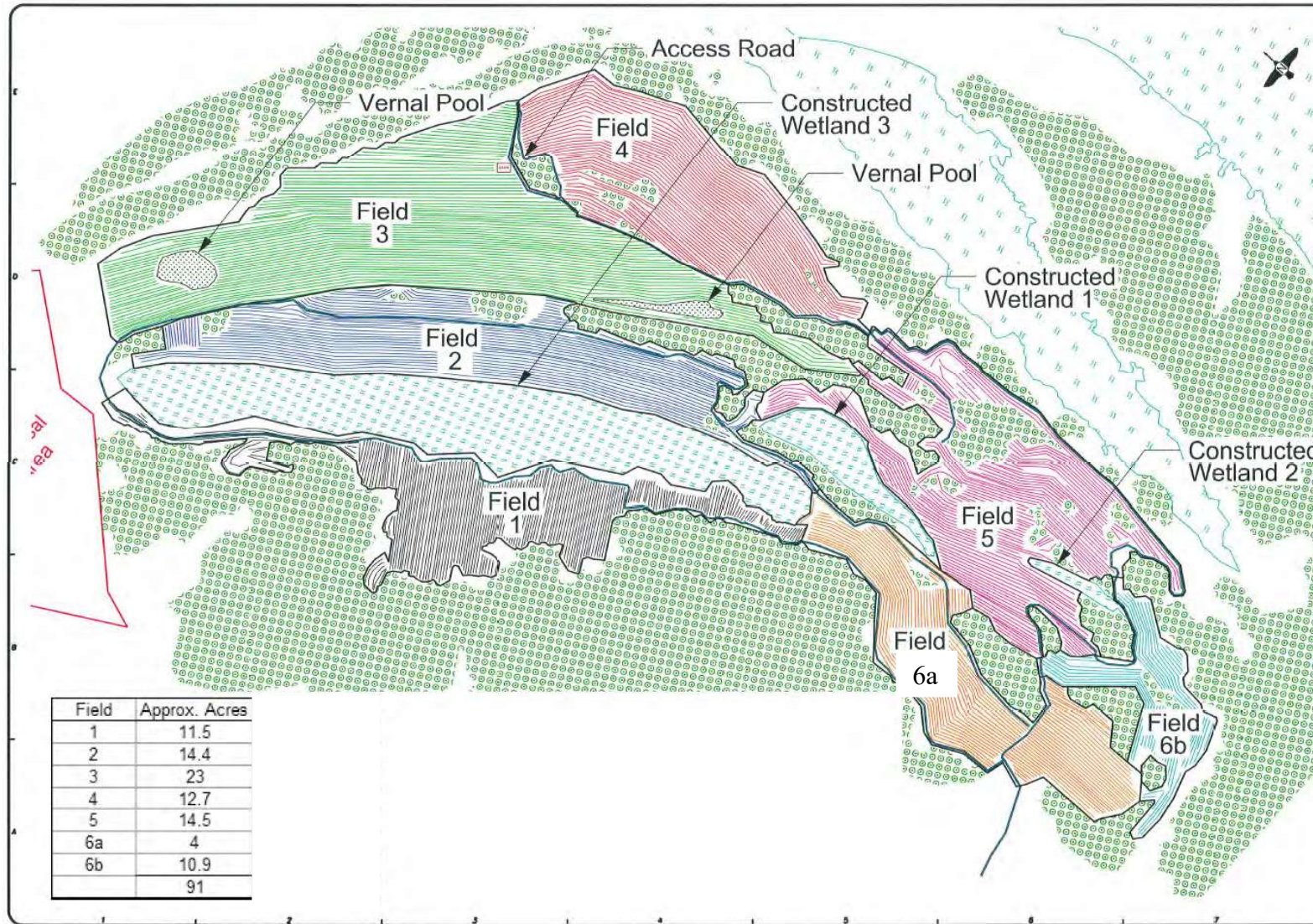
**Attachment 5:
Cottonwood Island Mitigation Site Location Map**



Attachment 6:
Cottonwood Island Mitigation Site Work Barge Landing and Road Access Map



**Attachment 7:
Cottonwood Island Mitigation Site As-Built Planting Map - Original Planting 2011**



Field	Approx. Acres
1	11.5
2	14.4
3	23
4	12.7
5	14.5
6a	4
6b	10.9
	91

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P.O. Box 350
Wenatchee, WA 98791
www.lkecorporation.com

USACE / LKE CORPORATION
Cottonwood Island Riparian Restoration
Area Map

WASHINGTON

PROJECT NUMBER
V91127N-09-C-0046

COTTONWOOD ISLAND
SCALE
NTS

Project Owner: US Army Corps of Engineers
Project Name: Cottonwood Island Riparian Restoration
Designed By: LKE
Drawn By: LKE
Reviewed By: LKE
Printed Date: 2-2-12

SHEET 03 OF 09
S-002

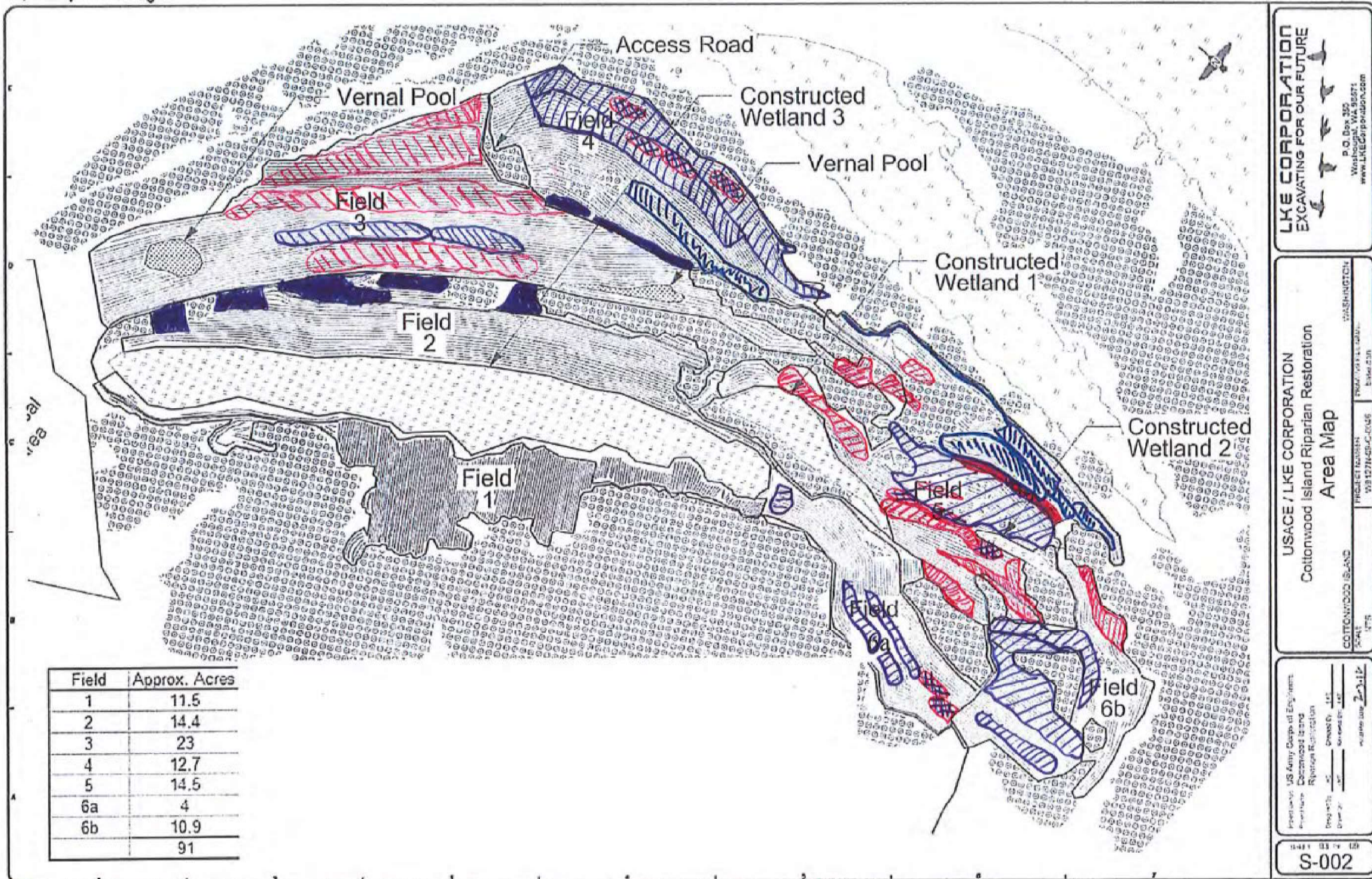
Attachment 8: Species of Cottonwood Island Original 2011 Plantings

SPECIFIC SPECIES – ESTIMATED QUANTITIES FROM INITIAL PLANTING (since this table was created, known 4,000 plantings have died and were not replaced – more plantings may have died since 2012)										
Field	Spacing Between Rows	Plant Spacing Along Row	Willow Tree	Cottonwood Tree	Red Osier	Douglas Spirea	Elderberry	Snowberry	Wild Rose	Total
1	10 ft	6 ft	4770	0	780	850	0	0	0	6400
2	10 ft	6 ft	6300	0	900	1050	0	0	0	8250
3	10 ft	6 ft	2500	8520	650	2300	1250	0	0	15220
4	10 ft	6 ft	0	6310	0	325	0	700	0	7335
5	10 ft	5 ft	600	5471	2000	50	0	400	0	8521
6a	10 ft	4.5 ft	50	1730	200	0	50	530	0	7960
6b	10 ft	3 ft	200	1548	1500	120	50	590	200	4208

Attachment 3:

Attachment 9: Cottonwood Island Mitigation Site As-Built 2011 Planting Map

1/2/2018



-  Cottonwoods installed Site #2 and site #3
-  Willows installed
-  Red osier Dogwood
-  Douglas Spirea
-  Snowberry

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P.O. Box 350
Washouak, WA 98671
www.LKEcorporation.com

USACE / LKE CORPORATION
Cottonwood Island Riparian Restoration
Area Map

PROJECT NUMBER: 13B-0214-01-0045
SCALE: 1"=100'
DATE: 1/2/18

Prepared by: US Army Corps of Engineers
Reviewed by: [Signature]
Date: 1/2/18

S-002

Attachment 10
Cottonwood Island Seed Areas

Legend

-  Cottonwood Mitigation Site Boundary
-  Seed Area





Attachment 11: Invasive Vegetation Classification

2025 CLARK COUNTY NOXIOUS WEED LIST



CLASS 'A'		CLASS 'B'			
ERADICATION REQUIRED		CONTROL OF CLASS 'B' & 'C' REGULATED WEEDS IS REQUIRED			
Eradication = destroy the entire plant and eliminate all plants		Control = prevent all seed production and prevent the dispersal of all propagative parts capable of forming new plants			
common crupina	<i>Crupina vulgaris</i>	blueweed	<i>Echium vulgare</i>	Ravenna grass	<i>Tripidium ravennae</i>
cordgrass, common	<i>Spartina anglica</i>	Brazilian elodea	<i>Egeria densa</i>	rush skeletonweed	<i>Chondrilla juncea</i>
cordgrass, dense-flowered	<i>Spartina densiflora</i>	bugloss, annual	<i>Lycopsis arvensis</i>	saltcedar	<i>Tamarix ramosissima</i>
cordgrass, saltmeadow	<i>Spartina patens</i>	bugloss, common	<i>Anchusa officinalis</i>	Scotch broom *	<i>Cytisus scoparius</i>
cordgrass, smooth	<i>Spartina alterniflora</i>	butterfly bush **	<i>Buddleja davidii</i>	shiny geranium *	<i>Geranium lucidum</i>
dyer's woad	<i>Isatis tinctoria</i>	camelthorn	<i>Alhagi maurorum</i>	spurge flax	<i>Thymelaea passerina</i>
eggleaf spurge	<i>Euphorbia oblongata</i>	common reed (nonnative)	<i>Phragmites australis</i>	spurge laurel	<i>Daphne laureola</i>
false brome	<i>Brachypodium sylvaticum</i>	common tansy	<i>Tanacetum vulgare</i>	spurge, leafy	<i>Euphorbia virgata</i>
floating primrose-willow	<i>Ludwigia peploides</i>	Dalmatian toadflax	<i>Linaria dalmatica ssp. dalmatica</i>	spurge, myrtle	<i>Euphorbia myrsinites</i>
flowering rush	<i>Butomus umbellatus</i>	Eurasian watermilfoil ***	<i>Myriophyllum spicatum</i>	sulfur cinquefoil	<i>Potentilla recta</i>
French broom	<i>Genista monspessulana</i>	European coltsfoot	<i>Tussilago farfara</i>	tansy ragwort	<i>Jacobaea vulgaris</i>
garlic mustard	<i>Alliaria petiolata</i>	fanwort	<i>Cabomba caroliniana</i>	thistle, musk	<i>Carduus nutans</i>
giant hogweed	<i>Heracleum mantegazzianum</i>	gorse	<i>Ulex europaeus</i>	thistle, plumeless	<i>Carduus acanthoides</i>
goatsrue	<i>Galega officinalis</i>	grass-leaved arrowhead	<i>Sagittaria graminea</i>	thistle, Scotch	<i>Onopordum acanthium</i>
hydrilla	<i>Hydrilla verticillata</i>	hairy willowherb	<i>Epilobium hirsutum</i>	velvetleaf	<i>Abutilon theophrasti</i>
Johnsongrass	<i>Sorghum halepense</i>	hawkweed oxtongue	<i>Picris hieracioides</i>	water primrose	<i>Ludwigia hexapetala</i>
knapweed, bighead	<i>Centaurea macrocephala</i>	hawkweed, orange	<i>Hieracium aurantiacum</i>	white bryony	<i>Bryonia alba</i>
knapweed, Vochin	<i>Centaurea nigrescens</i>	hawkweeds: all nonnative	<i>Hieracium, subgenus Pilosella</i>	wild chervil	<i>Anthriscus sylvestris</i>
kudzu	<i>Pueraria montana var. lobata</i>	hawkweeds: all nonnative	<i>Hieracium, subgenus Hieracium</i>	yellow floatingheart	<i>Nymphoides peltata</i>
meadow clary	<i>Salvia pratensis</i>	hoary alyssum	<i>Berteroa incana</i>	yellow nutsedge	<i>Cyperus esculentus</i>
oriental clematis	<i>Clematis orientalis</i>	houndstongue	<i>Cynoglossum officinale</i>	yellow starthistle	<i>Centaurea solstitialis</i>
Palmer amaranth	<i>Amaranthus palmeri</i>	knapweed, black	<i>Centaurea nigra</i>	CLASS 'C'	
purple starthistle	<i>Centaurea calcitrapa</i>	knapweed, brown	<i>Centaurea jacea</i>	absinth wormwood	<i>Artemisia absinthium</i>
reed sweetgrass	<i>Glyceria maxima</i>	knapweed, diffuse	<i>Centaurea diffusa</i>	babysbreath	<i>Gypsophila paniculata</i>
ricefield bulrush	<i>Schoenoplectus mucronatus</i>	knapweed, meadow *	<i>Centaurea x gerstlaueri</i>	black henbane	<i>Hyoscyamus niger</i>
roundlead bittersweet	<i>Celastrus orbiculatus</i>	knapweed, Russian	<i>Rhaponticum repens</i>	buffalobur	<i>Solanum rostratum</i>
sage, clary	<i>Salvia sclarea</i>	knapweed, spotted	<i>Centaurea stoebe</i>	fragrant waterlily	<i>Nymphaea odorata</i>
sage, Mediterranean	<i>Salvia aethiopis</i>	knotweed, Bohemian	<i>Fallopia x bohemica</i>	hairy whitetop	<i>Lepidium appelianum</i>
silverleaf nightshade	<i>Solanum elaeagnifolium</i>	knotweed, giant	<i>Fallopia sachalinensis</i>	hoary cress	<i>Lepidium draba</i>
small-flowered jewelweed	<i>Impatiens parviflora</i>	knotweed, Himalayan	<i>Persicaria wallichii</i>	medusahead	<i>Taeniatherum caput-medusae</i>
South American spongeplant	<i>Limnobium laevigatum</i>	knotweed, Japanese	<i>Fallopia japonica</i>	Russian olive	<i>Elaeagnus angustifolia</i>
Spanish broom	<i>Spartium junceum</i>	kochia	<i>Bassia scoparia</i>	spikeweed	<i>Centromadia pungens</i>
Syrian beancaper	<i>Zygophyllum fabago</i>	lesser celandine	<i>Ficaria verna</i>	thistle, bull *	<i>Cirsium vulgare</i>
Texas blueweed	<i>Helianthus ciliaris</i>	loosestrife, garden	<i>Lysimachia vulgaris</i>	thistle, Canada *	<i>Cirsium arvense</i>
thistle, Italian	<i>Carduus pycnocephalus</i>	loosestrife, purple	<i>Lythrum salicaria</i>	yellowflag iris	<i>Iris pseudacorus</i>
thistle, marsh	<i>Cirsium Palustre</i>	loosestrife, wand	<i>Lythrum virgatum</i>		
thistle, milk	<i>Silybum marianum</i>	Malta starthistle	<i>Centaurea melitensis</i>		
thistle, slenderflower	<i>Carduus tenuiflorus</i>	parrotfeather	<i>Myriophyllum aquaticum</i>		
thistle, Turkish	<i>Carduus cinereus</i>	poison hemlock	<i>Conium Maculatum</i>		
variable-leaf milfoil and hybrid	<i>Myriophyllum heterophyllum and M. heterophyllum x M. hippuroides</i>	policeman's helmet	<i>Impatiens glandulifera</i>		
wild four-o'clock	<i>Mirabilis nyctaginea</i>	puncturevine	<i>Tribulus terrestris</i>		

CLARK COUNTY NOXIOUS WEED CONTROL BOARD • 4700 NE 78TH STREET, BUILDING G - VANCOUVER, WA 98665 • (564) 397-6140 • WEED.BOARD@CLARK.WA.GOV

* Control is not mandated unless the weeds are within 100 feet of a neighboring property that is producing agriculture.

*** Control is mandated unless the weeds are within 100 feet of the Columbia River.

** Control is not mandated unless the weeds are within 100 feet of a natural waterbody.

Revised 02/05/25

<i>Tanacetum vulgare</i>	common tansy
<i>Thymelaea passerina</i>	spurge flax
<i>Tribulus terrestris</i>	puncturevine
<i>Trididum ravennae</i>	Ravenna grass
<i>Tussilago farfara</i>	European coltsfoot
<i>Ulex europaeus</i>	gorse

Class C Weeds

<i>Aegilops cylindrica</i>	jointed goatgrass
<i>Ailanthus altissima</i>	tree-of-heaven
<i>Alopecurus myosuroides</i>	blackgrass
<i>Artemisia absinthium</i>	absinth wormwood
<i>Arum italicum</i>	Italian arum
<i>Berberis vulgaris</i>	common barberry
<i>Cenchrus longispinus</i>	longspine sandbur
<i>Centromadia pungens</i>	spikeweed
<i>Cirsium arvense</i>	thistle, Canada
<i>Cirsium vulgare</i>	thistle, bull
<i>Clematis vitalba</i>	old man's beard
<i>Convolvulus arvensis</i>	field bindweed
<i>Cortaderia jubata</i>	jubata grass
<i>Cortaderia selloana</i>	Pampas grass
<i>Crataegus monogyna</i>	English hawthorn
<i>Cuscuta approximata</i>	smoothseed alfalfa dodder
<i>Daucus carota</i>	wild carrot (except where commercially grown)
<i>Dipsacus fullonum</i>	common teasel
<i>Elaeagnus angustifolia</i>	Russian olive
<i>Gypsophila paniculata</i>	babysbreath
<i>Hedera helix</i> 'Baltica', 'Pittsburgh', and 'Star', and <i>H. hibernica</i> 'Hibernica'	English ivy - four cultivars only
<i>Hyoscyamus niger</i>	black henbane
<i>Hypericum perforatum</i>	common St. Johnswort
<i>Hypochaeris radicata</i>	common catsear
<i>Impatiens capensis</i>	spotted jewelweed
<i>Iris pseudacorus</i>	yellow flag iris
<i>Lepidium appelianum</i>	hairy whitetop
<i>Lepidium draba</i>	hoary cress
<i>Leucanthemum vulgare</i>	oxeye daisy
<i>Linaria vulgaris</i>	yellow toadflax
<i>Myriophyllum spicatum</i> x <i>Myriophyllum sibiricum</i>	Eurasian watermilfoil hybrid
<i>Nanozostera japonica</i>	Japanese eelgrass
<i>Nymphaea odorata</i>	fragrant waterlily
<i>Pentaglottis sempervirens</i>	green alkanet

<i>Phalaris arundinacea</i>	reed canarygrass
<i>Potamogeton crispus</i>	curlyleaf pondweed
<i>Rorippa austriaca</i>	Austrian fieldcress
<i>Rubus bifrons</i> (<i>Rubus armeniacus</i>)	Himalayan blackberry
<i>Rubus laciniatus</i>	evergreen blackberry
<i>Secale cereale</i>	cereal rye
<i>Senecio vulgaris</i>	common groundsel
<i>Silene latifolia</i>	white cockle
<i>Solanum rostratum</i>	buffalobur
<i>Soliva sessilis</i>	lawnweed
<i>Sonchus arvensis</i>	perennial sowthistle
<i>Sphaerophysa salsula</i>	Swainsonpea
<i>Taeniatherum caput-medusae</i>	medusahead
<i>Tripleurospermum inodorum</i>	scentless mayweed
<i>Typha</i> species	nonnative cattail species and hybrids (reminder, does not include the native common cattail, <i>Typha latifolia</i>)
<i>Ventenata dubia</i>	ventenata
<i>Xanthium spinosum</i>	spiny cocklebur

To learn more about noxious weeds and noxious weed control in Washington State, please contact:

WA State Noxious Weed Control Board

P.O. Box 42560
Olympia, WA 98504-2560
(360) 725-5764

Email: noxiousweeds@agr.wa.gov

Website: <http://www.nwcb.wa.gov>

Or

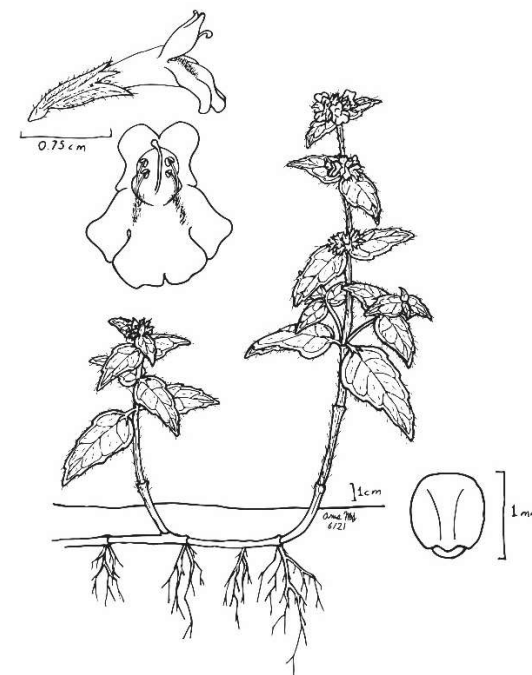
WA State Department of Agriculture
(509) 249-6973

Or

Your County Noxious Weed Control Board

2023

Washington State Noxious Weed List



Wild basil, *Clinopodium vulgare*, was a new Class B noxious weed for 2022. This invasive mint quickly invades disturbed areas and forest understories, creating monocultures.

Illustration by Anne Schuster

List arranged alphabetically by:
SCIENTIFIC NAME



ATTACHMENT 12:

CONTRACT DISCREPANCY REPORT

Instructions: QAP fills in blocks 1, 3, 4, 5 (prepared), and 6. Contractor fills in blocks 9 and 10.

1. CONTRACT NUMBER	2. REPORT NUMBER FOR THIS DISCREPANCY
3. TO: <i>(Contractor and Managers Name)</i>	4. FROM: <i>(Name of QAE)</i>

5. DATES		
PREPARED	RETURNED BY CONTRACTOR	ACTION COMPLETE

6. DISCREPANCY OR PROBLEM: *(Describe in detail: include applicable SOW/Contract references: attach continuation sheet if necessary.)*

7. SIGNATURE OF CONTRACTING OFFICER

8. TO: <i>(Contracting Officer)</i>	FROM: <i>(Contractor)</i>
-------------------------------------	---------------------------

9. CONTRACTOR RESPONSE AS TO CAUSE, CORRECTIVE ACTION AND ACTIONS TO PREVENT RECURRENCE: ATTACH CONTINUATION SHEET IF NECESSARY.
(Cite applicable Q.C. program procedures or new Q.C. procedures.)

10. SIGNATURE OF CONTRACTOR REPRESENTATIVE	DATE
--	------

11. GOVERNMENT EVALUATION *(Acceptance, partial acceptance, rejection; attach continuation sheet if necessary.)*

12. GOVERNMENT ACTIONS *(Reduced payment, cure notice, show cause, other.)*

CLOSE OUT			
	NAME-TITLE	SIGNATURE	DATE
CONTRACTOR NOTIFIED			
QAE			
ACO			

Attachment 13: Example GPS track screen shot at Chumbley



Attachment 14:

TECHNICAL MEMORANDUM

To: Jeanie Braun
Braun Construction
24805 SW Gage Road
Wilsonville, OR 97070

From: Chris Moller, Wetland Scientist / Project Manager

Date: August 6, 2021

Re: **Chumbley Wildlife Mitigation Site Vegetation Monitoring Plan / SWCA Project No. 67254**

PROJECT BACKGROUND

The Chumbley Wildlife Mitigation Site is located approximately 2 miles north of Woodland, Washington, adjacent to Interstate 5 and consists of 71 acres of monitoring area (Figure 1). The area consists of native species plantings which occurred at the site in 2009 (original plantings) and in 2018 (additional plantings to replace original failed plantings). Maintenance is required to control competing and invasive vegetation around the native plantings. The species of plantings and their approximate locations are shown in Attachments 3 and 4 in the RFP. Competing vegetation is defined as all vegetation within the site boundary except for plantings and native volunteers. The 2018 plantings are generally marked with plant protectors. Competing vegetation has grown taller than the plantings and protectors in many places. Invasive vegetation is defined as all Class A, Class B, and Class C plants listed in Attachment 11 in the RFP.

The contractor will distinguish on-site between plantings and other native species (Attachments 8 and 9) versus invasive species classes (Attachment 11) as outlined in the RFP. In 2019, mulch was applied in a layer 2 inches thick over the areas shown in Attachment 10 in the RFP (approximately 4 acres). Work under this contract will avoid disturbing the mulched areas.

PROJECT OVERVIEW

Monitoring Requirements

The contractor will conduct vegetation monitoring according to the methods in this vegetation monitoring plan. Two wetland scientists will record stem density, tree and shrub survival, invasive species cover, and native/desirable species at plot locations. Monitoring will occur over 2 days in the summer of 2021 (between June and September, depending on project schedule). Monitoring results will be presented in monitoring reports for each site. Requirements for monitoring and reporting are shown in Table 1.

Table 1. Riparian Planting Survival and Vegetation Assessment Requirements at Chumbley Wildlife Mitigation Site

Performance Standards	Fully Met? (Y/N)	Comments/Reason for shortfall (mark N/A if doesn't apply this year)
Establishment of Monitoring Transects and Plot Locations		
Required: 30 transect/plot locations must be established during the first annual monitoring in sufficient number and locations to be representative of the site.		
Required: Provide GPS coordinates for all transects/plots.		
Required: Establish photo points and GPS locations for each transect/plot location.		
Documenting Tree and Shrub Planting Survival (including re-plantings)		
Required: Stem (individual plant) counts of living planted and naturalized vegetation at transect/plot locations. Average of planting areas will be reported to assess tree and shrub survival.		
Measuring Invasive Species % Cover		
Required: Less than 20% invasive species cover across all strata. Percent cover measured across all plot locations.		
Measuring Native/Desirable Species % Cover		
Required: 80% or greater cover of native species across all vegetation strata. Percent cover measured across all plot locations.		

Note: GPS = global positioning system.

Monitoring Methods

The vegetation monitoring plan will establish a total of 30 monitoring plot locations established along five (5) transects. The transects will have six (6) sample randomly placed plots per transect. The transects run roughly north to south at equidistant intervals across the project area (Figure 2). The contractor will navigate to these plot locations, evaluate the area, and adjust as needed to avoid any built environment (buildings, roads, etc.) as well as wetlands and waterways where plantings did not occur. Final plot locations will be recorded by global positioning system (GPS) unit and provided for subsequent monitoring years.

Photographs of each sample plot will be taken to document conditions at time of monitoring. At each plot, two panoramic photos showing the entire area will be taken. The first panorama will be taken to the north (in an arc from west to east), and the second panorama will be taken to the south (in an arc from east to west). The photograph locations will be recorded at each sample plot location by GPS. GPS coordinates will be provided by the contractor to support subsequent year's monitoring activities.

At each plot location, the contractor's wetland scientists will conduct 15-foot-radius visual vegetation estimates to document species presence and percent cover of individual species within tree and scrub-shrub vegetation strata (Appendix A). Herbaceous and bare ground percent cover at each plot location will be evaluated within a 5-foot radius. Since vegetation within and between strata often overlaps there is the potential for more than 100% cover at sample plot locations. Visual vegetation estimation methodology will follow method prescribed in U.S. Army Corps of Engineers (USACE) *Wetlands Delineation Manual* (Environmental Laboratory 1987) and the *Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Western Mountains, Valleys, and Coast Region (Version 2.0)* (USACE 2010). In addition to recording the species and percent cover, stem counts of planted and naturalized shrub and tree species will be documented within a 15-foot radius.

Site access for Chumbley Monitoring Site will be coordinated with Braun Construction. Two days of monitoring field work are anticipated. Fieldwork will be scheduled once the monitoring plan has been finalized. In 2021, fieldwork should occur before August 15, 2021. However, in subsequent years, the contractor shall perform monitoring in May or June **and** before annual maintenance begins.

VEGETATION MONITORING REPORTING

Upon completion of field survey of the monitoring sites, the contractor will prepare the current-year monitoring report for the site. The report will detail maintenance activities, present vegetation data, evaluate results against performance standards, and recommend management actions. A draft report will be provided by November 1, and a final report will be provided by November 30, 2021. The report will include the following as relevant to the monitoring project:

- Project Name: Columbia River Channel Improvement Project Riparian Mitigation, Chumbley Mitigation Site.
- Name of party responsible for conducting the monitoring and the date(s) the inspection was conducted.
- A brief paragraph describing the purpose of the approved project, acreage and type of aquatic resources impacted, and mitigation acreage and type of aquatic resources authorized to compensate for the aquatic impacts.
- Written description of the location, any identifiable landmarks of the mitigation project including information to locate the site perimeter(s) and coordinates of the mitigation site (expressed as latitude/longitude, Universal Transverse Mercator, state plane coordinate system, etc.).
- Dates the mitigation project commenced and/or was completed.
- Short statement on whether the performance standards are being met.
- Dates of any recent corrective or maintenance activities conducted since the previous report submission.
- Specific recommendations for any additional corrective or remedial actions.

Requirements

List the monitoring requirements and performance standards as specified in the approved monitoring plan (see Table 1).

Monitoring Methods

Detailed description of the monitoring methods and rationale behind them.

Summary Data

Summary data in the form of descriptions and tables representing the monitoring data collected in the field will be provided. Field data will be assessed to determine the presence and percent cover of native/desirable species, the presence of invasive species (see Appendix A), and the stem counts and survival of planting within the monitoring sites. These data will be used assess the success and/or potential challenges associated with the mitigation project. Photo-documentation of the plots and site conditions will be provided to support the findings and recommendations referenced in the monitoring report.

Maps and Plans

Maps showing the project locations of the mitigation sites relative to other landscape features, habitat types, locations of photograph reference points, transects, sampling data points, and/or other features pertinent to the mitigation plan will be included. The maps will also include clearly delineated mitigation site perimeters.

CONCLUSIONS

The conclusions will include a description of the conditions of the mitigation project. If performance standards are not being met, the contractor will recommend remedial actions, including a brief explanation of the difficulties of remedial actions and a timetable for implementation.

LITERATURE CITED

- Environmental Laboratory. 1987. *Corps of Engineers Wetlands Delineation Manual*. Technical Report Y-87-1. Vicksburg, Mississippi: U.S. Department of the Army, Corps of Engineers, Waterways Experiment Station.
- U.S. Army Corps of Engineers (USACE). 2010. *Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Western Mountain, Valleys, and Coast Region (Version 2.0)*. edited by J.S. Wakeley, R.W. Lichvar, and C.V. Noble. ERDC/EL TR-10-3. Vicksburg, Mississippi: U.S. Army Corps of Engineers Engineer Research and Development Center.

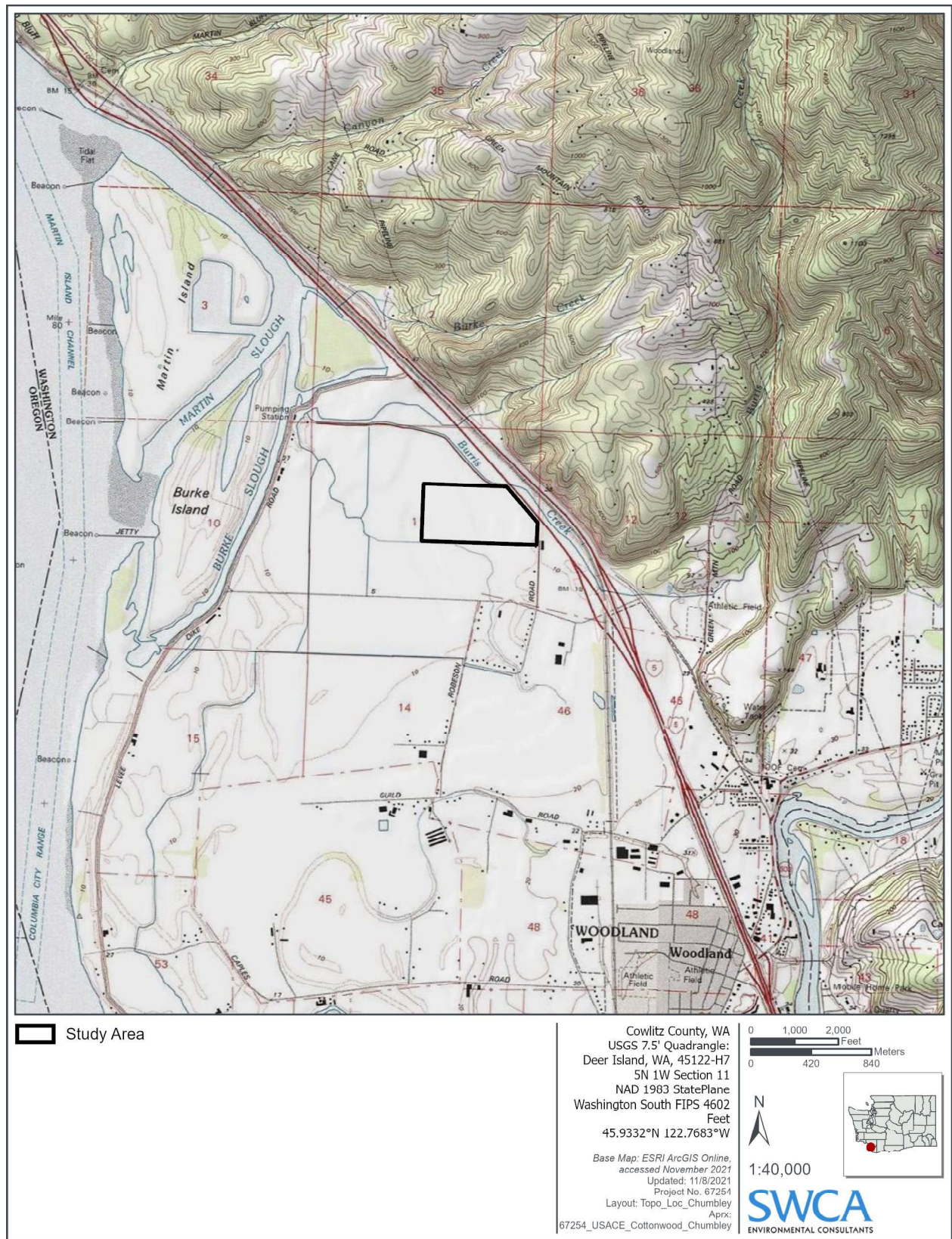


Figure 1. Chumbley Wildlife Mitigation Site location map. Shows the location of monitoring site north of Woodland, Washington (lower left).

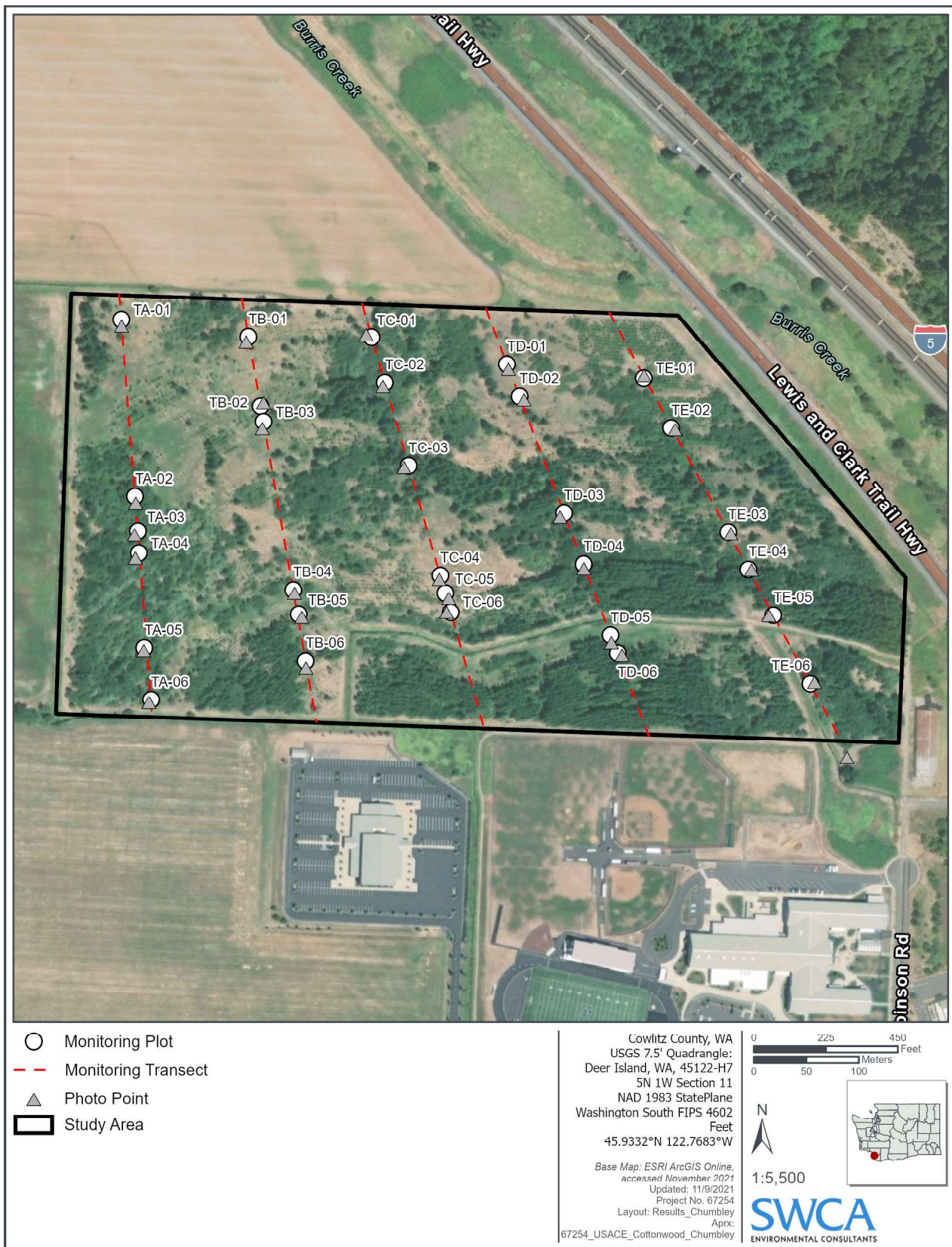


Figure 2. Chumbley Wildlife Mitigation Site sampling map. Shows the location of monitoring transect/plots relative to the tax parcels, mulched areas, planting zones, and study area boundary.

APPENDIX A

Monitoring Form Example

Instructions: Add or delete columns and rows to reflect the data collected. Review formulas in highlighted cells (green) and fix formulas as needed. After completing, use the Summary Information to evaluate site performance against the specific standards found in your monitoring plan. These standards and the Summary Information should be filled into the Monitoring Report Cover Sheet. Print this worksheet and include it in the monitoring report.

Site:																		
Upland Buffer Habitat Unit	Percent Cover																	
Species	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Row Average		
Native Herbaceous Species																		
<i>species-latin name</i>																		
Invasive Herbaceous Species																		
<i>species-latin name</i>																		
Non-Native Herbaceous Species																		
<i>species-latin name</i>																		
Native Shrub and Tree Species																		
<i>species-latin name</i>																		
Routine Performance Standards	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Habitat Average	Standard Error	
Cover of Native Herbaceous Species																#DIV/0!	#DIV/0!	
Lower CI (80%)																#DIV/0!		
Upper CI (80%)																#DIV/0!		
Cover of Invasive Herbaceous Species																#DIV/0!	#DIV/0!	
Lower CI (80%)																#DIV/0!		
Upper CI (80%)																#DIV/0!		
Cover of Native Shrubs and Trees																#DIV/0!	#DIV/0!	
Lower CI (80%)																#DIV/0!		
Upper CI (80%)																#DIV/0!		

Attachment 15: Chumbley Site Vegetation Monitoring Plot Locations

Table A-1. Vegetation Monitoring Point Coordinates (Plot Centers)

Monitoring Point Number	Latitude	Longitude
TA-01	45.934807	-122.77276
TA-02	45.933294	-122.77253
TA-03	45.93304	-122.77249
TA-04	45.932931	-122.77247
TA-05	45.932014	-122.77233
TA-06	45.931557	-122.77226
TB-01	45.934693	-122.77119
TB-02	45.934089	-122.77101
TB-03	45.933966	-122.77097
TB-04	45.932532	-122.77055
TB-05	45.932332	-122.77049
TB-06	45.931928	-122.77037
TC-01	45.934718	-122.76967
TC-02	45.934428	-122.76954
TC-03	45.933628	-122.76918
TC-04	45.932691	-122.76875
TC-05	45.932605	-122.76871
TC-06	45.932382	-122.76861
TD-01	45.934522	-122.768
TD-02	45.934249	-122.76784
TD-03	45.933346	-122.7673
TD-04	45.932826	-122.76699
TD-05	45.932238	-122.76664
TD-06	45.932082	-122.76655
TE-01	45.934427	-122.76634
TE-02	45.934042	-122.76598
TE-03	45.933056	-122.76516
TE-04	45.93282	-122.76496
TE-05	45.932435	-122.76464
TE-06	45.931859	-122.76417

Attachment 16: Chumbley Site Monitoring Photo Locations

Table A-2. Photo Point Coordinates

Photo Point Number	Latitude	Longitude
TA-01	45.934766	-122.77277
TA-02	45.933251	-122.77253
TA-03	45.93298	-122.77253
TA-04	45.93278	-122.77251
TA-05	45.931988	-122.77238
TA-06	45.931546	-122.7723
TB-01	45.934661	-122.77123
TB-02	45.934144	-122.771
TB-03	45.933921	-122.771
TB-04	45.93252	-122.77055
TB-05	45.932315	-122.77045
TB-06	45.931882	-122.77038
TC-01	45.934752	-122.76976
TC-02	45.934313	-122.76953
TC-03	45.933623	-122.76925
TC-04	45.932671	-122.76877
TC-05	45.932513	-122.76865
TC-06	45.932393	-122.76867
TD-01	45.934492	-122.768
TD-02	45.934233	-122.7678
TD-03	45.933238	-122.76731
TD-04	45.932799	-122.767
TD-05	45.932172	-122.76664
TD-06	45.932068	-122.7665
TE-01	45.93447	-122.76633
TE-02	45.934014	-122.76593
TE-03	45.933135	-122.76521
TE-04	45.932829	-122.76494
TE-05	45.932444	-122.76471
TE-06	45.931883	-122.76413

Attachment 17:



1800 NW Upshur Street, Suite 100
Portland, Oregon 97209
Tel 503.224.0333 Fax 503.224.0229
www.swca.com

TECHNICAL MEMORANDUM

To: Jeanie Braun
Braun Construction
24805 SW Gage Road
Wilsonville, OR 97070

From: Chris Moller, Wetland Scientist / Project Manager

Date: August 6, 2021

Re: **Cottonwood Island Wildlife Mitigation Site Vegetation Monitoring Plan / SWCA Project No. 67254**

PROJECT BACKGROUND

The Cottonwood Island Wildlife Mitigation Site is located on an island approximately 4.5 miles north of Kalama, Washington, and consists of 90 acres of monitoring areas (Figure 1). The site was originally planted with native species in 2011 (original plantings) and in 2018 (additional plantings to replace original plantings that failed). Maintenance is required to control competing and invasive vegetation around the 2011 mitigation plantings. The species of plantings are listed in Attachments 8 and 9 in the Request for Proposal (RFP). Competing vegetation is defined as all vegetation within the site boundary, except for plantings (and native volunteers). The plantings are generally marked with flags on metal stakes or plant protectors. Competing vegetation may have grown taller than the plantings and markers in many places, or the markers have been dislodged and only the metal stakes may remain. Invasive vegetation is defined as all Class A, Class B, and Class C plants listed in Attachment 11 in the RFP.

The contractor will distinguish on-site between plantings and other native species (Attachments 8 and 9) versus invasive species classes (Attachment 11) as outlined in the RFP. In 2019, mulch was applied in a layer 2 inches thick over the areas shown in Attachment 10 in the RFP (approximately 4 acres). Work under this contract will avoid disturbing the mulched areas.

PROJECT OVERVIEW

Monitoring Requirements

The contractor will conduct vegetation monitoring according to the methods in this vegetation monitoring plan. Two wetland scientists will record stem density, tree and shrub survival, invasive species cover, and native/desirable species at plot locations. Monitoring will occur over 2 days in the summer of 2021 (between June and September, depending on project schedule). Monitoring results will be presented in monitoring reports for each site. Requirements for monitoring and reporting are shown in Table 1.

Table 1. Riparian Planting Survival and Vegetation Assessment Requirements at Cottonwood Island Wildlife Mitigation Sites

Performance Standards	Fully Met? (Y/N)	Comments/Reason for shortfall (mark N/A if doesn't apply this year)
Establishment of Monitoring Transects and Plot Locations		
Required: 30 plot locations must be established during the first annual monitoring in sufficient number and locations to be representative of the site.		
Required: Provide GPS coordinates for all transects/plots.		
Required: Establish photo points and GPS locations for each transect/plot location.		
Documenting Tree and Shrub Planting Survival (including re-plantings)		
Required: Stem (individual plant) counts of living planted and naturalized vegetation at transect/plot locations. Average of planting areas will be reported and used to assess tree and shrub survival.		
Measuring Invasive Species % Cover		
Required: Less than 20% invasive species cover across all strata. Percent cover measured across all plot locations.		
Measuring Native/Desirable Species % Cover		
Required: 80% or greater cover of native species across all vegetation strata. Percent cover measured across all plot locations.		

Note: GPS = global positioning system.

Monitoring Methods

The vegetation monitoring plan will establish a total of 30 monitoring plot locations. The site is a mosaic of planting areas, wetlands, open water, mulched areas, and other treatment areas. Instead of transects, geographic information system (GIS) software was used to generate random locations within the planting areas. Plot locations were adjusted to avoid recently mulched areas (Figure 2) within the planting zones. The contractor will navigate to these plot locations, evaluate the area, and adjust as needed to avoid any built environment (buildings, roads, etc.) as well as wetlands and waterways where plantings did not occur. Final plot locations will be recorded by global positioning system (GPS) unit and provided for subsequent monitoring years.

Photographs of each sample plot will be taken to document conditions at time of monitoring. At each plot, two panoramic photos showing the entire area will be taken. The first panorama will be taken to the north (in an arc from west to east), and the second panorama will be taken to the south (in an arc from east to west). The photograph locations will be recorded at each sample plot location by GPS. Photographs of the eight (8) photograph locations from the 2013 As-Built Report will also be taken. GPS coordinates will be provided by the contractor to support subsequent year's monitoring activities.

At each plot location, the contractor's wetland scientists will conduct 15-foot-radius visual vegetation estimates to document species presence and percent cover of individual species within tree and scrub-shrub vegetation strata (Appendix A). Herbaceous and bare ground percent cover at each plot location will be evaluated within a 5-foot radius. Since vegetation within and between strata often overlaps there is the potential for more than 100% cover at sample plot locations. Visual vegetation estimation methodology will follow method prescribed in U.S. Army Corps of Engineers (USACE) *Wetlands Delineation Manual* (Environmental Laboratory 1987) and the *Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Western Mountains, Valleys, and Coast Region (Version 2.0)*

(USACE 2010). In addition to recording the species and percent cover, stem counts of planted and naturalized shrub and tree species will be documented within a 15-foot radius.

Transportation and site access for the Cottonwood Island Wildlife Mitigation Site will be coordinated with Braun Construction. An access map to planting areas is provided in Attachment 6 in the RFP. Two days of monitoring field work are anticipated. Fieldwork will be scheduled once the monitoring plan has been finalized. In 2021, fieldwork should occur before August 15, 2021. However, in subsequent years, the contractor shall perform monitoring in May or June **and** before annual maintenance begins.

VEGETATION MONITORING REPORTING

Upon completion of field survey of the monitoring sites, the contractor will prepare the current-year monitoring report for the site. The report will detail maintenance activities, present vegetation data, evaluate results against performance standards, and recommend management actions. A draft report will be provided by November 1, and a final report will be provided by November 15, 2021. The report will include the following as relevant to the monitoring project:

- Project Name: Columbia River Channel Improvement Project Riparian Mitigation, Cottonwood Island Mitigation Site.
- Name of party responsible for conducting the monitoring and the date(s) the inspection was conducted.
- A brief paragraph describing the purpose of the approved project, acreage and type of aquatic resources impacted, and mitigation acreage and type of aquatic resources authorized to compensate for the aquatic impacts.
- Written description of the location, any identifiable landmarks of the mitigation project including information to locate the site perimeter(s) and coordinates of the mitigation site (expressed as latitude/longitude, Universal Transverse Mercator, state plane coordinate system, etc.).
- Dates the mitigation project commenced and/or was completed.
- Short statement on whether the performance standards are being met.
- Dates of any recent corrective or maintenance activities conducted since the previous report submission.
- Specific recommendations for any additional corrective or remedial actions.

Requirements

List the monitoring requirements and performance standards as specified in the approved monitoring plan (see Table 1).

Monitoring Methods

Detailed description of the monitoring methods and rationale behind them.

Summary Data

Summary data in the form of descriptions and tables representing the monitoring data collected in the field will be provided. Field data will be assessed to determine the presence and percent cover of native/desirable species, the presence of invasive species (see Appendix A), and the stem counts and survival of planting within the monitoring sites. These data will be used assess the success and/or

potential challenges associated with the mitigation project. Photo-documentation of the plots and site conditions will be provided to support the findings and recommendations referenced in the monitoring report.

Maps and Plans

Maps showing the project locations of the mitigation sites relative to other landscape features, habitat types, locations of photograph reference points, transects, sampling data points, and/or other features pertinent to the mitigation plan will be included. The maps will also include clearly delineated mitigation site perimeters.

CONCLUSIONS

The conclusions will include a description of the conditions of the mitigation project. If performance standards are not being met, the contractor will recommend remedial actions, including a brief explanation of the difficulties of remedial actions and a timetable for implementation.

LITERATURE CITED

- Environmental Laboratory. 1987. *Corps of Engineers Wetlands Delineation Manual*. Technical Report Y-87-1. Vicksburg, Mississippi: U.S. Department of the Army, Corps of Engineers, Waterways Experiment Station.
- U.S. Army Corps of Engineers (USACE). 2010. *Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Western Mountain, Valleys, and Coast Region (Version 2.0)*. edited by J.S. Wakeley, R.W. Lichvar, and C.V. Noble. ERDC/EL TR-10-3. Vicksburg, Mississippi: U.S. Army Corps of Engineers Engineer Research and Development Center.

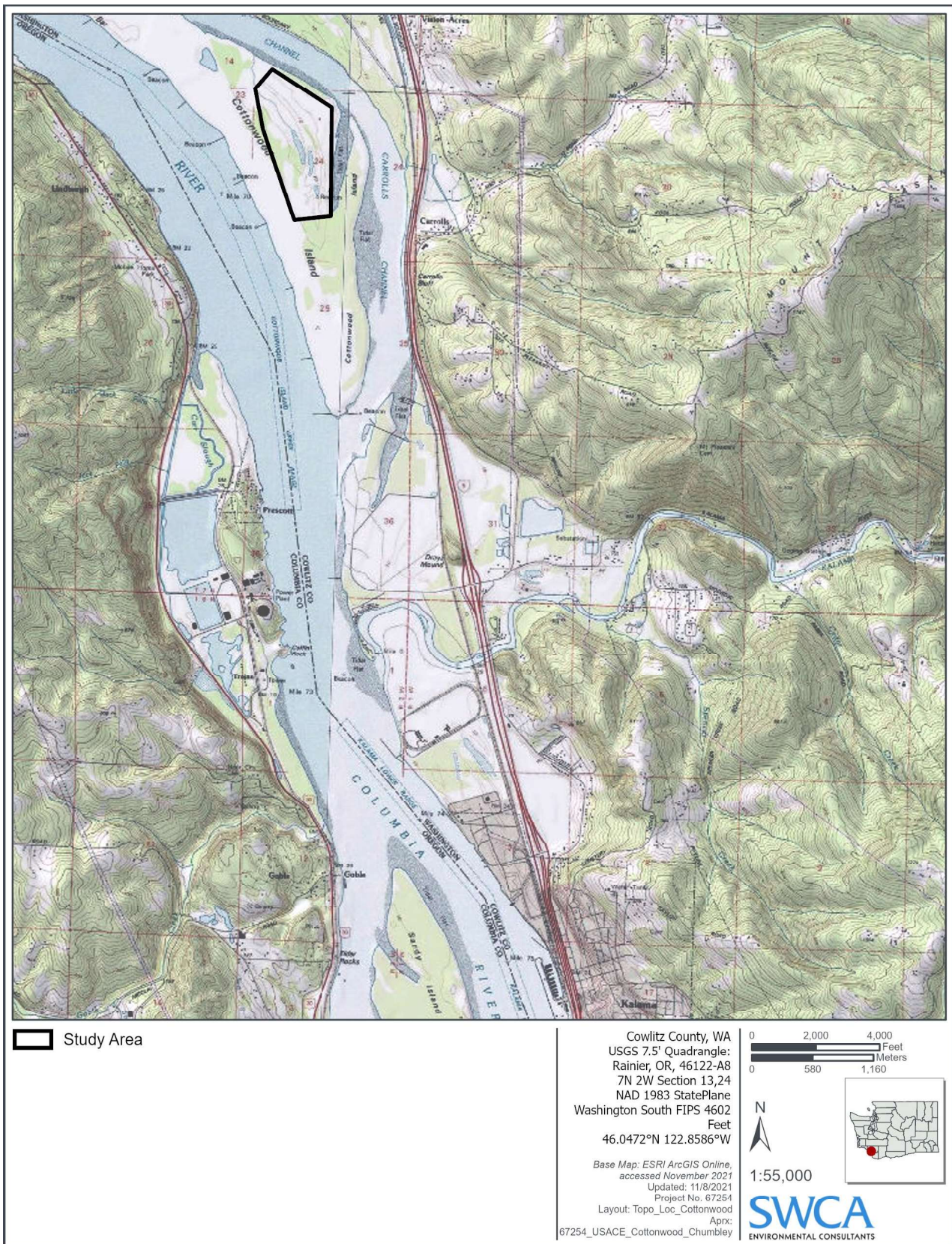


Figure 1. Cottonwood Island Wildlife Mitigation Site location map. Shows the location of monitoring site north of Kalama, Washington (lower left).



Figure 2. Cottonwood Island Wildlife Mitigation Site sampling map. Shows the location of monitoring transect/plots relative to the tax parcels, mulched areas, planting zones, and study area boundary.

APPENDIX A

Monitoring Form Example

Instructions: Add or delete columns and rows to reflect the data collected. Review formulas in highlighted cells (green) and fix formulas as needed. After completing, use the Summary Information to evaluate site performance against the specific standards found in your monitoring plan. These standards and the Summary Information should be filled into the Monitoring Report Cover Sheet. Print this worksheet and include it in the monitoring report.

Site:																			
Upland Buffer Habitat Unit	Percent Cover																		
Species	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Row Average			
Native Herbaceous Species																			
<i>species-latin name</i>																			
Invasive Herbaceous Species																			
<i>species-latin name</i>																			
Non-Native Herbaceous Species																			
<i>species-latin name</i>																			
Native Shrub and Tree Species																			
<i>species-latin name</i>																			
Routine Performance Standards	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Habitat Average	Standard Error		
Cover of Native Herbaceous Species																#DIV/0!	#DIV/0!		
Lower CI (80%)																#DIV/0!			
Upper CI (80%)																#DIV/0!			
Cover of Invasive Herbaceous Species																#DIV/0!	#DIV/0!		
Lower CI (80%)																#DIV/0!			
Upper CI (80%)																#DIV/0!			
Cover of Native Shrubs and Trees																#DIV/0!	#DIV/0!		
Lower CI (80%)																#DIV/0!			
Upper CI (80%)																#DIV/0!			

Attachment 18: Cottonwood Island Vegetation Monitoring Plot Locations

Table A-1. Vegetation Monitoring Plots, Coordinates of Plot Centers

Monitoring Point Number	Latitude	Longitude
1	46.073662	-122.87936
2	46.073844	-122.88049
3	46.072959	-122.88176
4	46.075342	-122.88186
5	46.076798	-122.88142
6	46.07376	-122.88108
7	46.076376	-122.87968
8	46.076831	-122.87997
9	46.075089	-122.87918
10	46.075417	-122.87959
11	46.075729	-122.87859
12	46.081921	-122.87981
13	46.078441	-122.87907
14	46.080781	-122.87862
15	46.079812	-122.87898
16	46.079644	-122.88482
17	46.07908	-122.88387
18	46.08021	-122.88519
19	46.078682	-122.88313
20	46.079025	-122.88121
21	46.083144	-122.88576
22	46.082795	-122.88501
23	46.079922	-122.88221
24	46.081934	-122.88346
25	46.081688	-122.88104
26	46.081603	-122.88214
27	46.083384	-122.8843
28	46.080641	-122.88045
29	46.082953	-122.88371
30	46.082762	-122.88161

Attachment 19: Cottonwood Island Monitoring Photo Locations

Table A-2. Photo Point Coordinates

Photo Point Number	Latitude	Longitude
1	46.073628	-122.8794
2	46.073821	-122.88044
3	46.072929	-122.88183
4	46.075291	-122.88186
5	46.076762	-122.88142
6	46.073704	-122.88107
7	46.076324	-122.87969
8	46.076784	-122.87994
9	46.075029	-122.8792
10	46.075359	-122.87956
11	46.075675	-122.87862
12	46.081903	-122.87975
13	46.0784	-122.87905
14	46.080718	-122.87863
15	46.079767	-122.87896
16	46.079608	-122.8848
17	46.07905	-122.88385
18	46.080164	-122.88519
19	46.078697	-122.8831
20	46.078972	-122.8812
21	46.083123	-122.8857
22	46.08277	-122.88497
23	46.079885	-122.88224
24	46.081902	-122.88342
25	46.081647	-122.88099
26	46.081569	-122.88209
27	46.083345	-122.88425
28	46.080591	-122.88045
29	46.082922	-122.88366
30	46.082714	-122.88157

Attachment 20: Cottonwood Island Reference Photo Locations

Table A-3. 2013 As-Built Reference Point Photo Coordinates.

Reference Point Number	Coordinates (NAD83 Oregon North, US survey feet)	Coordinates (WGS 1984, Decimal Degrees)	Description
1	7597714, 886575	46.073935, -122.881588	Southeastern portion of mitigation site, field 6, near southeast entrance of island.
2	7597909, 886728	46.07437, -122.880838	Follows constructed path, southeast portion of site, north of reference point 1, junction of field 6 and field 5.
3	7597896, 887037	46.075213, -122.880927	Southeastern portion of site, west of reference point 2, northern junction of fields 5 and 6.
4	7597925, 888123	46.078194, -122.880939	Central portion of site, junction between fields 5 and 2, borders northern portion of constructed wetland 1.
5	7597722, 888969	46.080496, -122.881834	Western portion of site, junction between central portions of fields 3 and 2, north of constructed wetland 3.
6	7597312, 888282	46.078579, -122.883369	Southwestern portion of site, eastern portion of field 1, south of constructed wetland 3.
7	7596610, 889547	46.081989, -122.886283	Western portion of site, western portion of field 1, south of constructed wetland 3.
8	7596561, 890218	46.083826, -122.886554	Western portion of site, western portion of field 3, junction between fields 3 and 1 and constructed wetland 3.

Attachment 21:

TECHNICAL MEMORANDUM

To: Jeanie Braun
Braun Construction
24805 SW Gage Road
Wilsonville, OR 97070

From: Chris Moller, Wetland Scientist / Project Manager

Date: June 28, 2022

Re: **Chumbley and Cottonwood Wildlife Mitigation Site Wildlife Usage Monitoring Plan / SWCA Project No. 67254**

PROJECT BACKGROUND

The Chumbley Wildlife Mitigation Site is located approximately 2 miles north of Woodland, Washington, adjacent to Interstate 5 and consists of 71 acres of monitoring area. The area consists of native species plantings which occurred at the site in 2009 (original plantings) and in 2018 (additional plantings to replace original failed plantings). The Cottonwood Island Wildlife Mitigation Site is located on an island approximately 4.5 miles north of Kalama, Washington, and consists of 90 acres of monitoring areas. The site was originally planted with native species in 2011 (original plantings) and in 2018 (additional plantings to replace original plantings that failed). Maintenance is required to control competing and invasive vegetation around the 2011 mitigation plantings.

PROJECT OVERVIEW

Monitoring Requirements

The contractor will conduct monitoring according to the methods in this wildlife usage monitoring plan. One wildlife biologist will conduct the monitoring at the site. Monitoring will occur over 2 days in the summer of 2021 (between July and September, depending on project schedule). Monitoring results will be presented in monitoring reports for each site.

Background

Monitoring the health and long-term stability of bird communities can provide an important measure of overall environmental health (Smits and Fernie 2013) and effectiveness of management decisions (Noson and Hutto 2005). Birds are good environmental monitors for several reasons: many species can be monitored simultaneously with a single method, methods for monitoring are well understood and standardized, birds occupy all habitat types, and birds as a community represent several trophic levels and habitat-use guilds. Monitoring species abundance, community diversity, and trends provides information that can be used to determine the effectiveness of management actions in moving towards restoration goals.

Point counts are the most widely used quantitative method for monitoring land birds and involve an observer recording birds from a single point for a standardized time period (Ralph et al. 1995). The methodology follows recommendations by Ralph et al. (1995) and is consistent with that employed by the Forest Service Northern Region Land Bird Monitoring Project (Hutto et al. 2001) and recommendations for the Idaho Partners in Flight Bird Monitoring Plan (Leukering et al. 2000). Fixed-radius plots (where the radius is arbitrarily small) reduce the interspecific difference in detectability by assuming that: 1) all the birds within the fixed-radius are detectable; 2) observers do not actively attract or repel birds; and 3) birds do not move into or out of the fixed-radius plot during the counting period. This allows for comparisons of abundance among species. Unlimited radius plots maximize the amount of data collected because they include all detections and are appropriate when the objective is to monitor population changes within a single population (Ralph et al. 1995). Birds should be tallied in two distance bands, one 0 to 50 m from the point center and one more than 50 m from the point center. This maximizes data collection while permitting interspecific analysis. Additional information on establishing point count stations, data collection, and sample data forms can be found in Ralph et al. (1993), Ralph et al. (1995), and Huff et al. (2000).

Protocol

Bird populations are sampled by the point-count method. Each sampling point is the center of a point-count station. The focal survey area consists of a 50-m radius circle around each point. All point-count stations will be georeferenced and recorded on the point-count data form. All birds detected are recorded and specified as being within or outside the focal survey area. Bird surveys are conducted from July to late August beginning at approximately 0700 hours and ending by 1100 hours. A single observer spends 8 minutes at each birding station recording in three intervals (0–3 minutes, 3–5 minutes, and 5–8 minutes). All birds seen or heard within each of these three time periods are recorded. Species are identified using a four-letter code based on common name (e.g., RWBL for red-winged blackbird). A generic code is used if the species cannot be identified. Birds encountered are assigned a group type from among the following: male (M), female (F), pair (P), nest (N), flock (FL), or unknown (U). An observation type is also assigned: singing (S), calling (C), visual (V), or drumming (D). For flocks, the approximate number of individuals is noted. Birds that are observed flying within the plot, but do not appear to be using the area, are noted as flyovers.

Weather conditions can have a great influence on the effectiveness of a survey. Upon arrival at each monitoring station, weather conditions are noted as: clear (C), partly cloudy (P), overcast (O), drizzle (D), rain (R), or fog (F). Because most birds are observed by sound, wind or rain can mask songs or call notes enough that they are not discernible to the observer. High wind and heavy rain can also force high canopy foragers to take shelter or generally decrease the morning activity of most birds. Surveys are not conducted, or are discontinued, if these weather conditions exist.

All stations should be visited two times during the breeding season (late May to late August, though late May to early July is ideal), with a minimum of 1 week between counts. Exact timing of sampling may vary due to conditions outside the observers' control (e.g., road and weather conditions). Subsequent analyses are weighted by the number of entries. To the extent possible, the order of daily visits to the point-count stations is reversed for each entry to increase the probability of observing both early and late morning singers across the point-count stations. To maximize the probability of recording all bird species present at a station regardless of variable arrival and breeding times, surveys are scheduled so that each station is visited throughout the breeding season. Observations while moving between survey points of notable species or nesting behaviors, or of species not detected during point count surveys within a mitigation property, are recorded in field notes and reported as incidental sightings.

WILDLIFE USAGE MONITORING REPORTING

Upon completion of field survey of the monitoring sites, the contractor will prepare a summary of wildlife use at the sites and include it in the vegetation monitoring report. A draft report will be provided by November 1, and a final report will be provided by November 30. The summary will include the following as relevant to the monitoring project:

- Name of party responsible for conducting the monitoring and the date(s) the inspection was conducted.
- Short statement on whether the performance standards are being met.

Maps and Plans

Maps showing the project locations of the mitigation sites relative to other landscape features, habitat types, locations of photograph reference points, point count locations, and specific siting details.

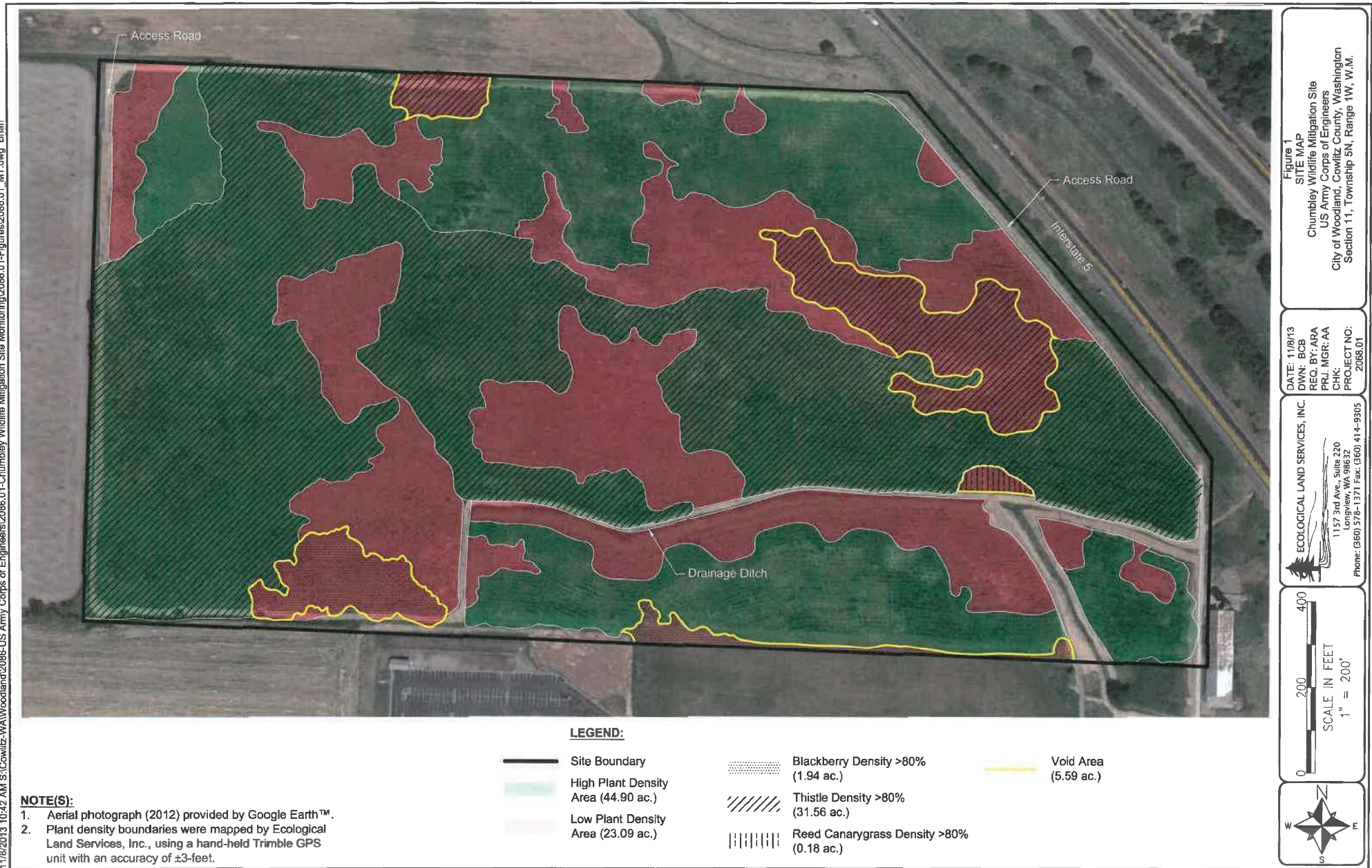
LITERATURE CITED

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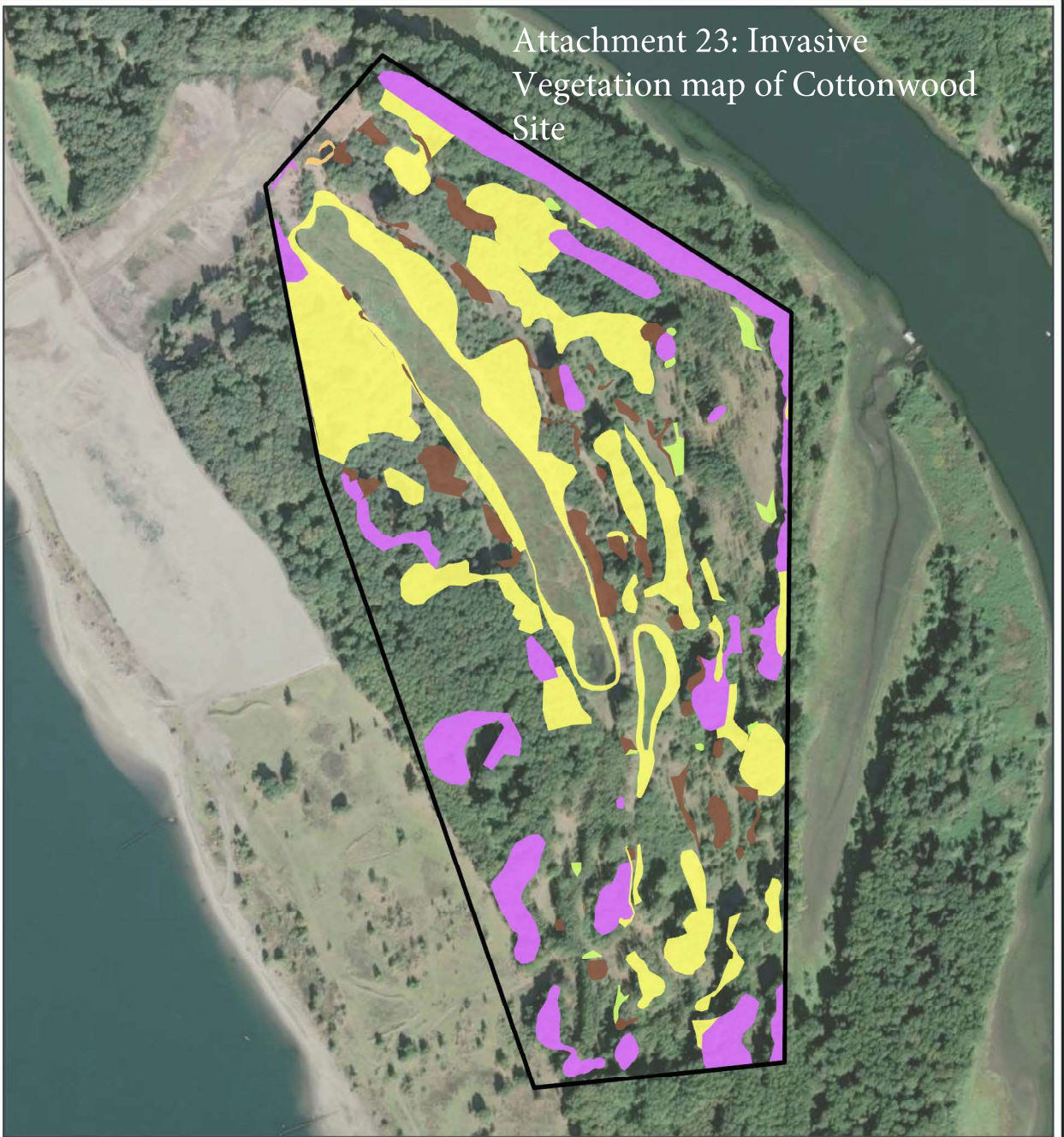
APPENDIX A

Monitoring Form Example

Attachment 22: Example Walk-Through Survey Results Map



Attachment 23: Invasive Vegetation map of Cottonwood Site



Mitigation Site

Species

Canada Thistle (*Cirsium arvense*) – 6.77 acres
 Himalayan Blackberry (*Rubus armeniacus*) – 19.36 acres

Reed Canary Grass (*Phalaris arundinacea*) – 37.80 acres
 Scotch Broom (*Cytisus scoparius*) – 0.12 acre
 Tansy Ragwort (*Jacobaea vulgaris*) – 1.14 acres

Cowlitz County, WA
 USGS 7.5' Quadrangle:
 Rainier, OR, 46122-A8
 7N 2W Section 13,24
 NAD 1983 StatePlane
 Washington South FIPS 4602
 Feet
 46.0786°N 122.8828°W
 Base Map: Esri ArcGIS Online,
 accessed September 2025
 Updated: 9/26/2025
 Project No. 67254
 Layout:
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 Aprx:
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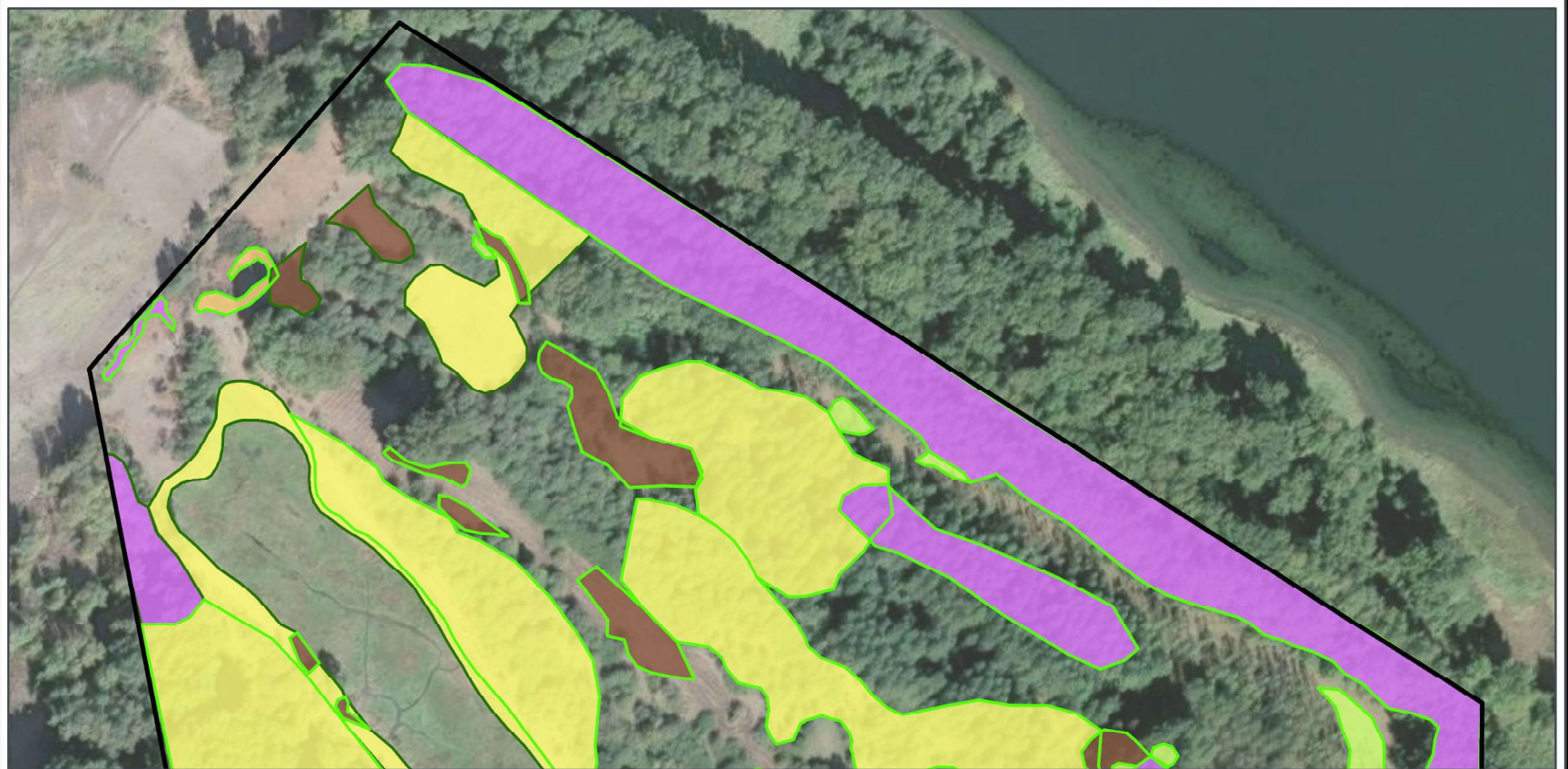
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SWCA
 ENVIRONMENTAL CONSULTANTS



Mitigation Site

Class B Invasive Species

Tansy Ragwort
(*Jacobaea vulgaris*)

Class C Invasive Species

Canada Thistle (*Cirsium arvense*)

Himalayan Blackberry
(*Rubus armeniacus*)

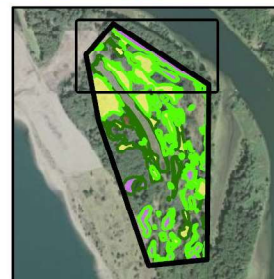
Reed Canary Grass
(*Phalaris arundinacea*)

Scotch Broom (*Cytisus scoparius*)

Percent Cover

30%-50% Cover

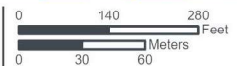
≥80% Cover



Cowlitz County, WA
USGS 7.5' Quadrangle:
Rainier, OR, 46122-A8
7N 2W Section 13,24
NAD 1983 StatePlane
Washington South FIPS 4602
Feet
46.0834°N 122.8831°W

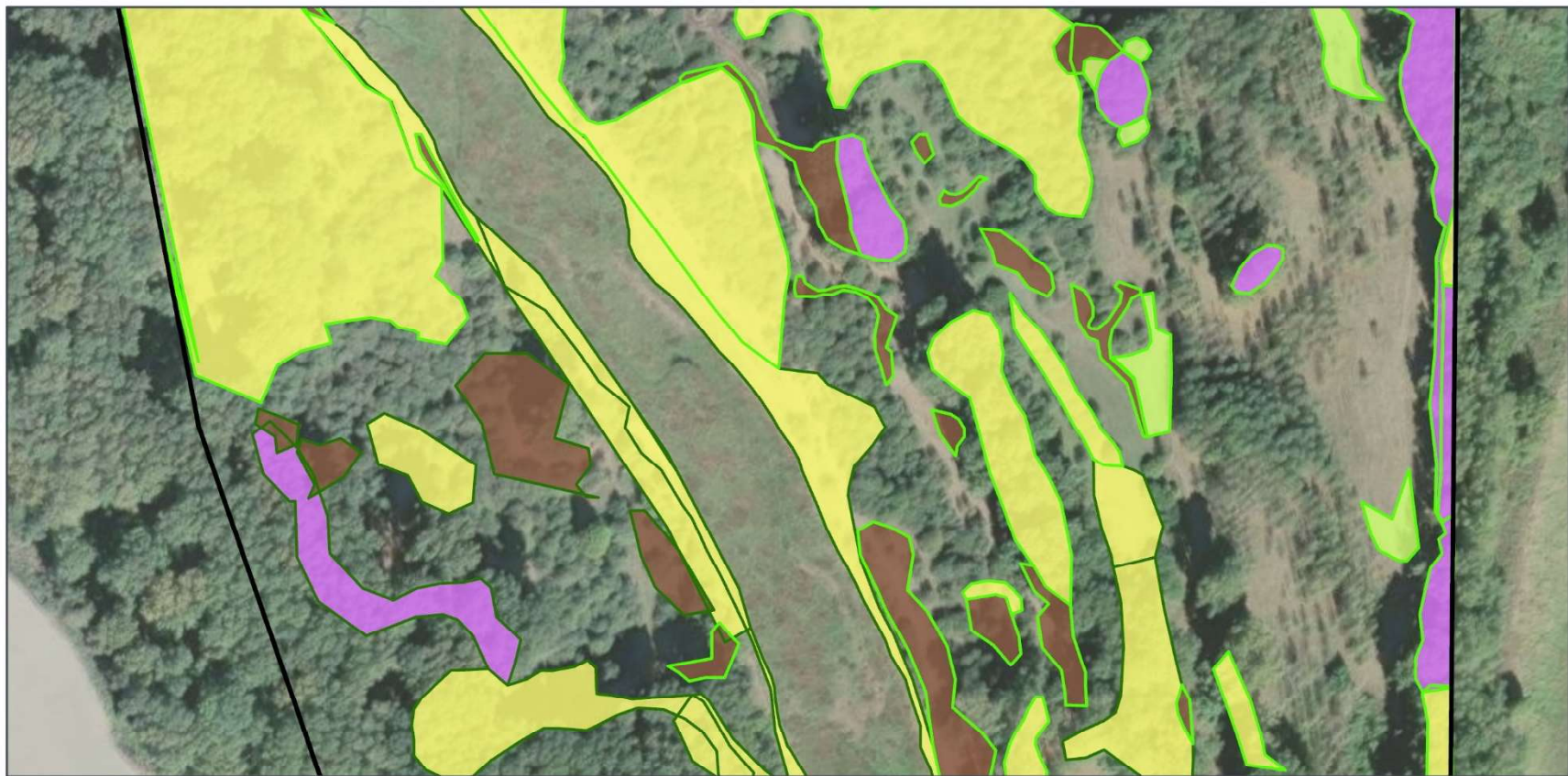
Base Map: Esri ArcGIS Online,
accessed October 2025
Updated: 10/14/2025
Project No. 82406

Layout:
InvasiveVegetation_Cottonwood
Aprx:
98942_CottonwoodChumbley_2025



1:3,600

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Mitigation Site

Class B Invasive Species

Tansy Ragwort
(*Jacobaea vulgaris*)

Class C Invasive Species

Canada Thistle (*Cirsium arvense*)

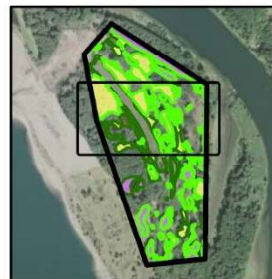
Himalayan Blackberry
(*Rubus armeniacus*)

Reed Canary Grass
(*Phalaris arundinacea*)

Percent Cover

30%-50% Cover

≥80% Cover



Cowlitz County, WA
USGS 7.5' Quadrangle:
Rainier, OR, 46122-A8
7N 2W Section 13,24
NAD 1983 StatePlane
Washington South FIPS 4602
Feet
46.08°N 122.8827°W

Base Map: Esri ArcGIS Online,
accessed October 2025
Updated: 10/14/2025
Project No. 82406
Layout:
InvasiveVegetation_Cottonwood
Aprx:
98942_CottonwoodChumbley_2025

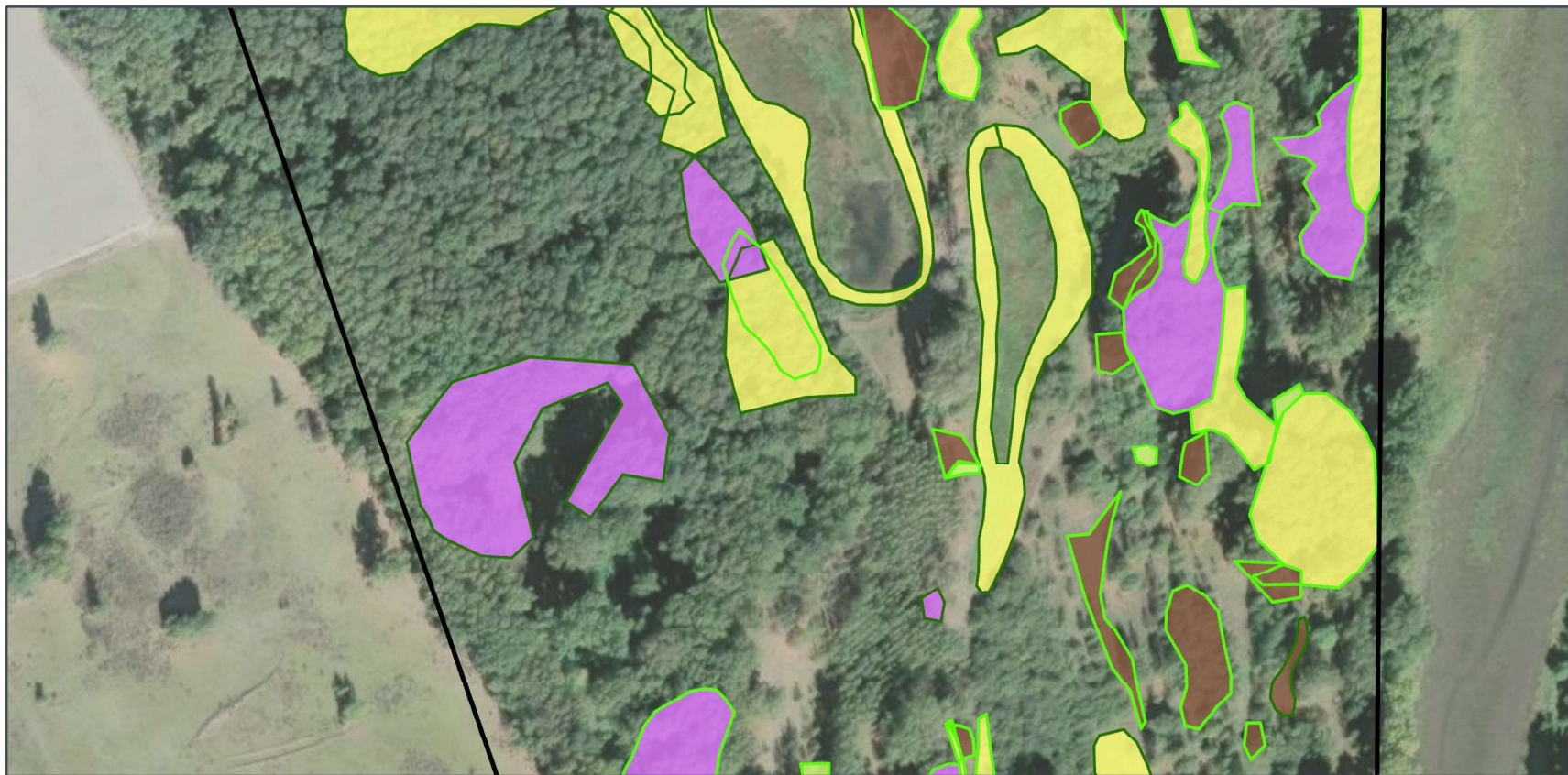
0 140 280 Feet
0 30 60 Meters



1:3,600



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Mitigation Site

Class B Invasive Species

Tansy Ragwort
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Percent Cover

30%-50% Cover

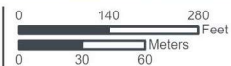
≥80% Cover



Cowlitz County, WA
USGS 7.5' Quadrangle:
Rainier, OR, 46122-A8
7N 2W Section 13,24
NAD 1983 StatePlane
Washington South FIPS 4602
Feet
46.0768°N 122.8821°W

Base Map: Esri ArcGIS Online,
accessed October 2025
Updated: 10/14/2025
Project No. 82406

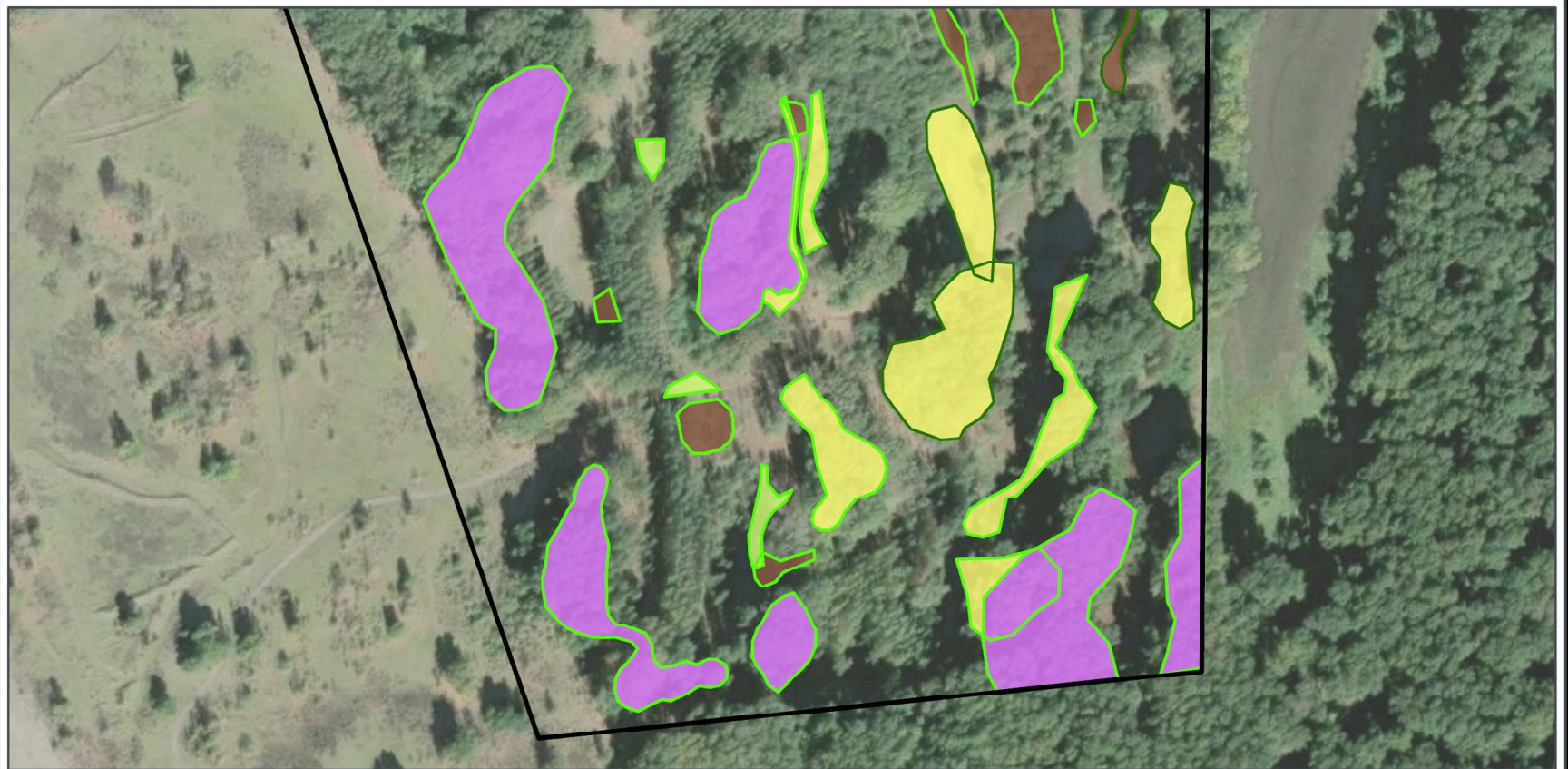
Layout:
InvasiveVegetation_Cottonwood
Aprx:
98942_CottonwoodChumbley_2025



1:3,600



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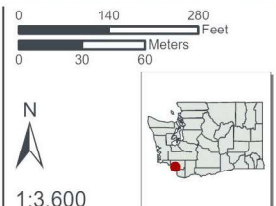


- Mitigation Site
Class B Invasive Species
 Tansy Ragwort (*Jacobaea vulgaris*)
Class C Invasive Species
 Canada Thistle (*Cirsium arvense*)
- Himalayan Blackberry (*Rubus armeniacus*)
 Reed Canary Grass (*Phalaris arundinacea*)
- Percent Cover**
 30%-50% Cover
 ≥80% Cover



Cowlitz County, WA
 USGS 7.5' Quadrangle:
 Rainier, OR, 46122-A8
 7N 2W Section 13,24
 NAD 1983 StatePlane
 Washington South FIPS 4602
 Feet
 46.0739°N 122.8809°W

Base Map: Esri ArcGIS Online,
 accessed October 2025
 Updated: 10/14/2025
 Project No. 82406
 Layout:
 InvasiveVegetation_Cottonwood
 Aprx:
 98942_CottonwoodChumbley_2025



1:3,600

SWCA
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